# Membrane flutter in three-dimensional inviscid flow
# Movie Caption List

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**Caption for movie 1:**
We show 3 types of membrane motion from Group 1 in the order mentioned below.

i. Membrane motion with (FRFF, fixed-free-fixed-fixed) boundary conditions and with dimensionless quantities: membrane mass $R_1 = 10^{-1}$, pretension $T_0 = 10^{-0.75}$, and stretching rigidity $R_3 = 10^1$. This motion corresponds to the purple region ( ) in figure 12.

ii. Membrane motion with (FRFR, fixed-free-fixed-free) boundary conditions and with dimensionless quantities: membrane mass $R_1 = 10^1$, pretension $T_0 = 10^{-0.75}$, and stretching rigidity $R_3 = 10^1$. This motion corresponds to the green region ( ) in figure 12.

iii. Membrane motion with (FFFF, fixed-fixed-fixed-fixed) boundary conditions and with dimensionless quantities: membrane mass $R_1 = 10^0$, pretension $T_0 = 10^{-0.25}$, and stretching rigidity $R_3 = 10^0$. This motion corresponds to the yellow region ( ) in figure 12 and it is periodic.

**Caption for movie 2:**
We show 3 types of membrane motion from Group 2 in the order mentioned below.

i. Membrane motion with (FRRF, fixed-free-free-fixed) boundary conditions and with dimensionless quantities: membrane mass $R_1 = 10^{-1}$, pretension $T_0 = 10^{-1.25}$, and stretching rigidity $R_3 = 10^0$. This motion corresponds to the small-mass region of figure 13 and it is periodic.

ii. Membrane motion with (FRRF, fixed-free-free-fixed) boundary conditions and with dimensionless quantities: membrane mass $R_1 = 10^{0.5}$, pretension $T_0 = 10^{-0.5}$, and stretching rigidity $R_3 = 10^1$. This motion corresponds to the moderate-mass region of figure 13 and it is periodic.

iii. Membrane motion with (FRRR, fixed-free-free-free) boundary conditions and with dimensionless quantities: membrane mass $R_1 = 10^{1.5}$, pretension $T_0 = 10^{-0.5}$, and stretching rigidity $R_3 = 10^1$. This motion corresponds to the large-mass region of figure 13.

**Caption for movie 3:**
Group 3: Membrane motion with (RRFF, free-free-fixed-fixed) boundary conditions and with dimensionless quantities: membrane mass $R_1 = 10^{-1}$, pretension $T_0 = 10^{1}$, and stretching rigidity $R_3 = 10^{4}$. This motion corresponds to the light purple region ( ) in figure 14 and it is periodic.

**Caption for movie 4:**
Group 3: Membrane motion with (RFFF, free-fixed-fixed-fixed) boundary conditions and with dimensionless quantities: membrane mass $R_1 = 10^{-1.5}$, pretension $T_0 = 10^{0.5}$, and stretching rigidity $R_3 = 10^{0}$. This motion corresponds to the yellow region ( ) in figure 14 and it is periodic.

**Caption for movie 5:**
Group 3: Membrane motion with (RRFR, free-free-fixed-free) boundary conditions and with dimensionless quantities: membrane mass $R_1 = 10^{-1.5}$, pretension $T_0 = 10^{1.75}$, and stretching rigidity $R_3 = 10^{1}$. This motion corresponds to the dark purple region ( ) in figure 14 and it is periodic.

**Caption for movie 6:**
We show 3 types of membrane motion from Group 4 in the order mentioned below.

i. Membrane motion with (RRRR, free-free-free-free) boundary conditions and with dimensionless quantities: membrane mass $R_1 = 10^{0.5}$, pretension $T_0 = 10^{-0.5}$, and stretching rigidity $R_3 = 10^{1}$. This motion corresponds to the gray region ( ) in figure 15.

ii. Membrane motion with (RRRF, free-free-free-fixed) boundary conditions and with dimensionless quantities: membrane mass $R_1 = 10^{-0.5}$, pretension $T_0 = 10^{-0.75}$, and stretching rigidity $R_3 = 10^{1}$. This motion corresponds to the purple region ( ) in figure 15.

iii. Membrane motion with (RFRF, free-fixed-free-fixed) boundary conditions and with dimensionless quantities: membrane mass $R_1 = 10^{-2}$, pretension $T_0 = 10^{-0.75}$, and stretching rigidity $R_3 = 10^{1}$. This motion corresponds to the blue region ( ) in figure 15.

**Caption for movie 7:**
Group 4: Membrane motion with (RFRF, free-fixed-free-fixed) boundary conditions and with dimensionless quantities: membrane mass $R_1 = 10^{-0.5}$, pretension $T_0 = 10^{-1}$, and stretching rigidity $R_3 = 10^{1}$. This motion corresponds to the green region ( ) in figure 15 and it is periodic.

**Caption for movie 8:**
Group 4: Membrane motion with (RFRF, free-fixed-free-fixed) boundary conditions and with dimensionless quantities: membrane mass $R_1 = 10^{2}$, pretension $T_0 = 10^{-0.5}$, and stretching rigidity $R_3 = 10^{1}$. This motion corresponds to the yellow region ( ) in figure 15.

**Caption for movie 9:**
Group 4: Membrane motion with (RRRR, free-free-free-free) boundary conditions and with dimensionless quantities: membrane mass $R_1 = 10^0$, pretension $T_0 = 10^{-0.75}$, and stretching rigidity $R_3 = 10^1$. This motion corresponds to the orange region ( ) in figure 15 and it is periodic.

**Caption for movie 10:**
Group 4: Membrane motion with (RRRR, free-free-free-free) boundary conditions and with dimensionless quantities: membrane mass $R_1 = 10^{-2}$, pretension $T_0 = 10^{-1}$, and stretching rigidity $R_3 = 10^1$. This motion corresponds to the red region ( ) in figure 15.

The $x$-axis shows the $x(\alpha_1, \alpha_2, t)$-coordinate of the membrane, the $y$-axis shows the $y(\alpha_1, \alpha_2, t)$-coordinate, and the $z$-axis shows the $z(\alpha_1, \alpha_2, t)$-coordinate. In all the videos the frame rate has been adjusted to show the motion over a time period that captures the overall motion of the membrane.